

ABSTRACT & MOTIVATION

Policy Knowledge Graphs (KGs) are crucial for decision support (compliance, analysis, QA) and enhancing Large Language Models (LLMs). Existing research lacks methods for constructing *fine-grained* policy KGs, especially for specialized domains like forestry. **Problem:** Forestry policy texts contain long, abstract entities and complex relationships, posing challenges for traditional and even direct LLM-based extraction. Labeled data is scarce. **Our Contribution:**

- A fine-grained **forestry policy domain ontology**.
- An **unsupervised policy information extraction method** leveraging open-source LLMs (Kimi).
- The first open-source **Forestry Policy Knowledge Graph (ForPKG)**.
- Demonstrated practical value via **Retrieval-Augmented Generation**.

FINE-GRAINED ONTOLOGY DESIGN**10 Entity Types:**

- General: ORG, PER, LOC
- Policy-Specific: DOC (Policy Document), CLS (Categories)
- Domain Concepts: CONC, OBJ, EXP_DEF
- Deontic Logic-Inspired: ACT, STATE

15 Relationship Types:

- Examples: publish, locate, belongTo, define & Deontic Logic-Based: haveTheDuty, prohibit

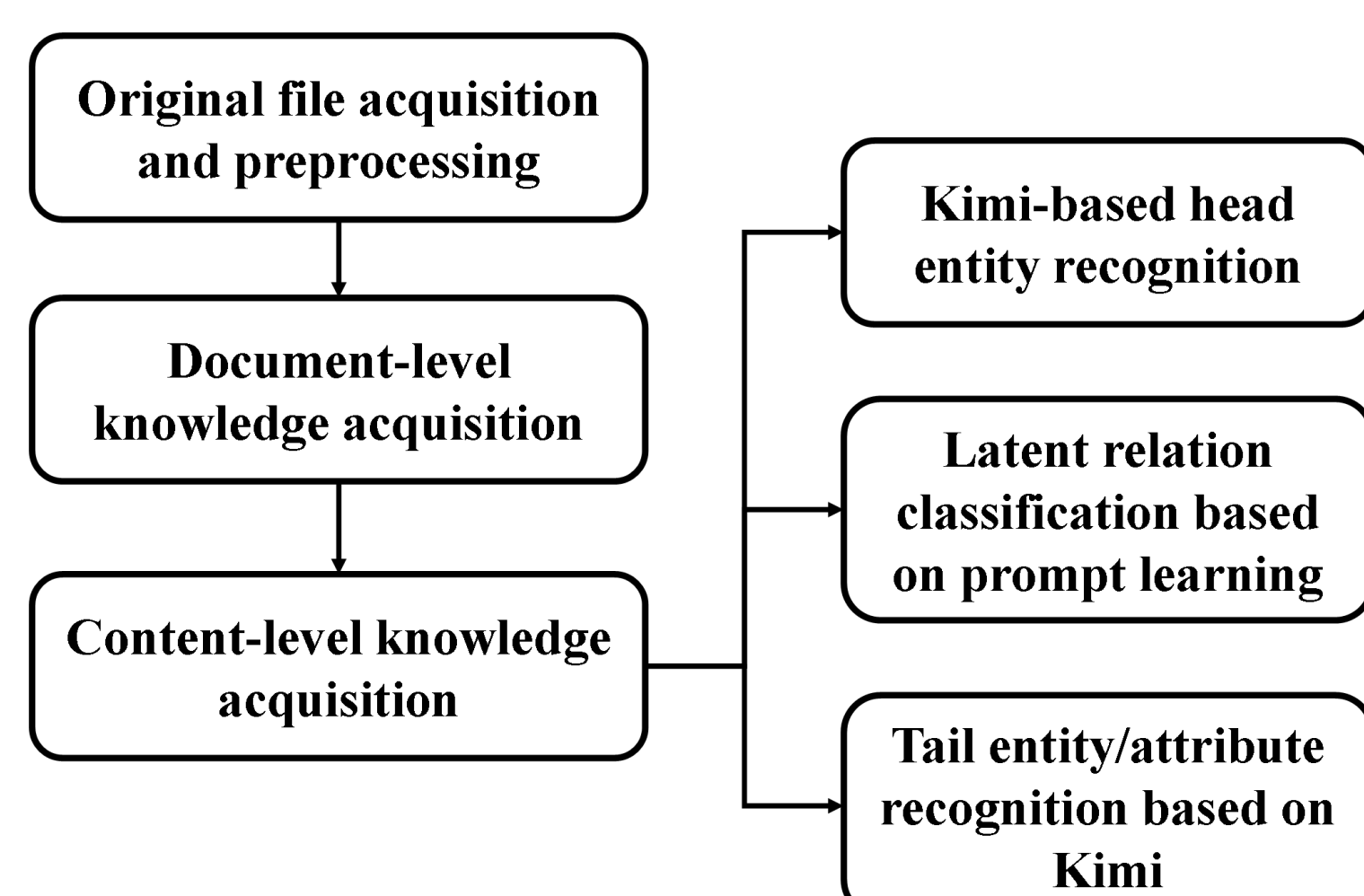
CONSTRUCTION FRAMEWORK

Figure: Workflow for building ForPKG using an open-source LLM.

Steps:

1. Original File Acquisition & Preprocessing (757 documents)
2. Document-Level Knowledge Acquisition (Metadata, BigBird for similarity)
3. Content-Level Knowledge Acquisition (LLM-based Triples)
 1. Kimi-based Head Entity Recognition;
 2. Latent Relation Classification (Prompt Learning);
 3. Tail Entity and Attribute Recognition (Kimi).

LLM-BASED INFORMATION EXTRACTION

Rationale: Overcomes labeled data scarcity using open-source LLM (Kimi). **Three-Step Process for Content-Level Extraction:**

1. **Head Entity Recognition (Kimi):** Identifies shorter, simpler head entities first. Full text input for context.
2. **Latent Relation Classification (Prompt Learning):** Model: Fine-tuned DeBERTa. Data: Few-shot (10 samples/15 relations = 150 total).
 $\hat{y} = M_{\text{DeBERTa}}(\text{prompt}(\text{text}, \text{head_entity}))$
3. **Tail Entity Recognition (Kimi):** Identifies relationship phrase and subsequent tail entity.

Addresses long entities and abstract relationships better than direct LLM prompting.

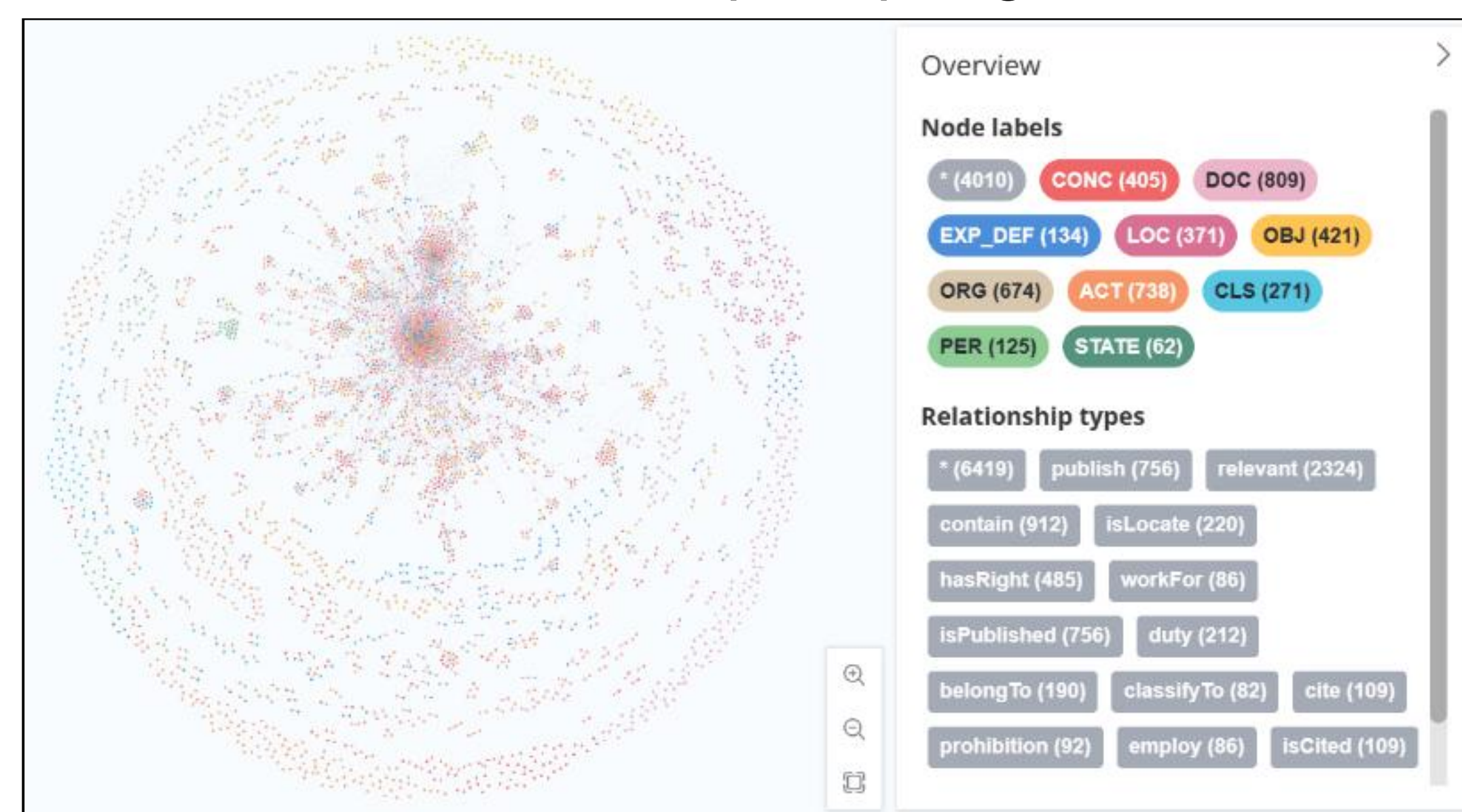


Figure: ForPKG visualization in Neo4j. (Nodes: 6.8k, Rels: 6.4k)

EXPERIMENTAL RESULTS: EXTRACTION PERFORMANCE

To evaluate triple extraction, we used 50 manually annotated forestry policy documents (1,126 triads). Table III details the precision and recall of our method against baselines.

Table: Precision and Recall of Triple Extraction

Method	Precision	Recall
<i>A ready-to-use unsupervised information extraction framework</i>		
DeepKE	22.3%	5.2%
LTP	16.5%	2.2%
Jiagu	32.9%	3.6%
DeepDive	19.1%	1.9%
<i>Calling the LLM API directly</i>		
Kimi- paragraph-level extraction	43.2%	17.9%
Kimi- document-level extraction	50.7%	23.3%
LLaMa2-Chinese- paragraph-level extraction	38.6%	12.8%
LLaMa2-Chinese- document-level extraction	30.7%	9.8%
Our method (ForPKG)	76.2%	62.6%

Key Findings from Table:

- **ForPKG (76.2% P / 62.6% R)** significantly outperforms: Unsupervised frameworks (e.g., DeepKE: 22.3% P 5.2% R; Jiagu: 32.9% P 3.6% R). Direct LLM API calls (e.g., Kimi- document-level: 50.7% P 23.3% R).
- This highlights ForPKG's enhanced capability in extracting policy-specific long/complex entities and abstract relationships.

EXPERIMENTAL RESULTS: ONTOLOGY APPLICATION

Ontology Scalability Flexibility: Fusing ForPKG with 5 general KGs (OwnThink, OpenKG, etc.) significantly improved query response accuracy (e.g., +3.4% with OwnThink).

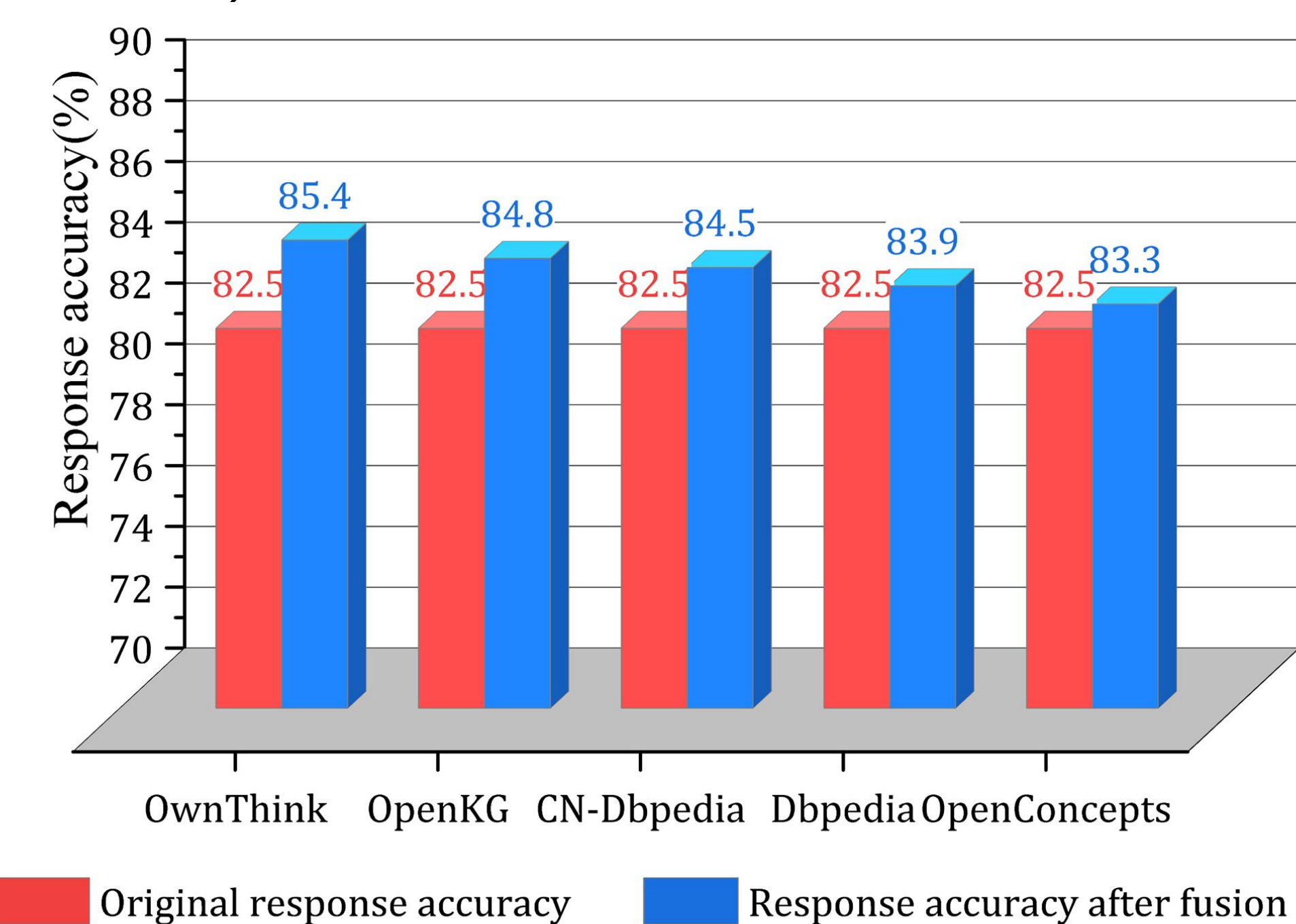


Figure: Response Accuracy Improvement After Fusion.

Practical Application: RAG for LLMs

ForPKG used as external knowledge for LLaMa-Chinese. Evaluation by GPT-4. **Result:** Significant improvement in LLaMa-Chinese generation quality.

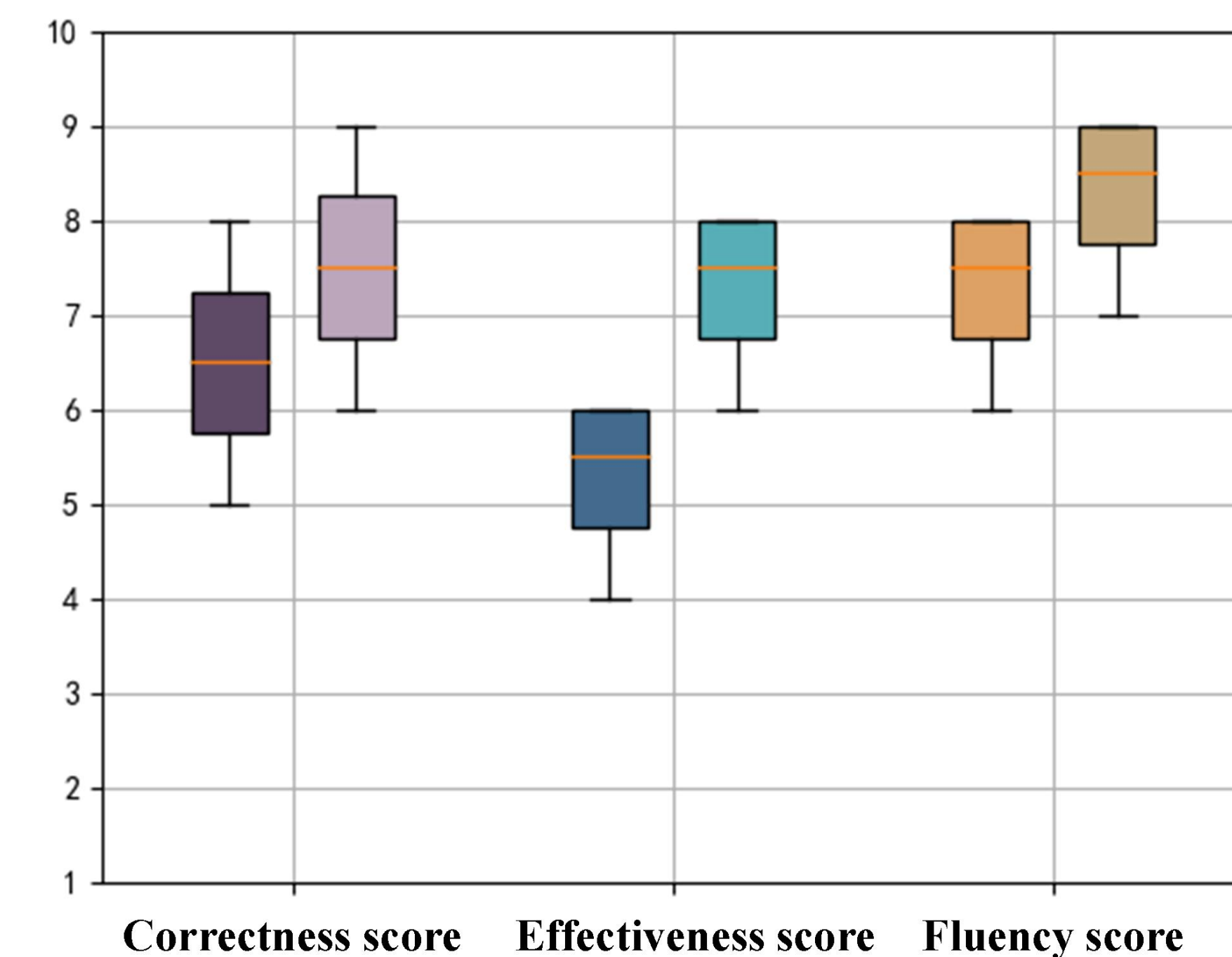


Figure: RAG Performance Improvement with ForPKG.

CONCLUSION FUTURE WORK**Conclusion:**

- ForPKG: A complete and effective framework for fine-grained policy KGs.
- Novel Ontology: Our design accurately captures complex forestry policy nuances.
- LLM-based Extraction: Unsupervised method proven effective for intricate policy texts.
- Practical Value: ForPKG demonstrates significant utility, especially in RAG for LLM.

Future Work:

- Further scale up ForPKG by incorporating more diverse documents.
- Refine ontology and extraction techniques for even greater accuracy and coverage.
- Develop an accessible online API for broader research and practitioner access.